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A time-dependent discrimination index for survival data

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SUMMARY

To derive models suitable for outcome prediction, a crucial aspect is the availability of appropriate measures of predictive accuracy, which have to be usable for a general class of models. The Harrell's C discrimination index is an extension of the area under the ROC curve to the case of censored survival data, which owns a straightforward interpretability. For a model including covariates with time-dependent effects and/or time-dependent covariates, the original definition of C would require the prediction of individual failure times, which is not generally addressed in most clinical applications. Here we propose a time-dependent discrimination index C^{td} where the whole predicted survival function is utilized as outcome prediction, and the ability to discriminate among subjects having different outcome is summarized over time. C^{td} is based on a novel definition of concordance: a subject who developed the event should have a less predicted probability of surviving beyond his/her survival time than any subject who survived longer. The predicted survival function of a subject who developed the event is compared to: (1) that of subjects who developed the event before his/her survival time, and (2) that of subjects who developed the event, or were censored, after his/her survival time. Subjects who were censored are involved in comparisons with subjects who developed the event before their observed times. The index reduces to the previous C in the presence of separation between survival curves on the whole follow-up. A confidence interval for C^{td} is derived using the jackknife method on correlated one-sample U-statistics.

The proposed index is used to evaluate the discrimination ability of a model, including covariates having time-dependent effects, concerning time to relapse in breast cancer patients treated with adjuvant tamoxifen. The model was obtained from 596 patients entered prospectively at Istituto Nazionale per lo Studio e la Cura dei Tumori di Milano (INT). The model discrimination ability was validated on an independent testing data set of 175 patients provided by Centro Regionale Indicatori Biochimici di Tumore (CRIBT) in Venice. Copyright © 2005 John Wiley & Sons, Ltd.

KEY WORDS: discrimination; time dependence; survival model; ROC analysis; non-parametric method

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1. INTRODUCTION

Multivariable regression models are widely used in biomedical research with the aim of predicting the outcome of individual patients [1]. In this context, the assessment of the model performance has to focus on the accuracy of the predictions, rather than merely on the covariate effects and their statistical significance. Similarly, when new covariates are available, their possible contribution to the model has to be evaluated by the gain in predictive accuracy [2, 3].

A basic requirement a model has to satisfy to be used for predictive purposes is generalizability, i.e. the ability to provide accurate predictions for new patients [1]. Generalizability can be considered at two levels: to patients coming from the same population where the model was developed (reproducibility), and to patients coming from a different plausibly related population (transportability) [4]. Assessing the predictive accuracy 'naively' on the model training data set, leads to an overoptimistic (biased) result. The evaluation of the degree of optimism of the result, and the subsequent bias corrected predictive accuracy, may offer insights into the 'true' model reproducibility. However, to directly face this issue, the predictive accuracy has to be evaluated resorting to independent testing data sets [4].

Several approaches to derive measures of predictive accuracy have been proposed: explained variation, for the extent to which the covariates determine the individual outcome [3, 5, 6]; accuracy scores, for the quantitative agreement between predicted and observed outcome [7, 8]; indexes of separation between predicted survival curves specific for prognostic strata [9]; discrimination, for the ability of the outcome predictions in separating subjects consistently to the observed outcome [10]. Here we focus on the last approach, given its straightforward interpretability from the clinical viewpoint.

A discrimination measure for survival models is the C index, proposed by Harrell *et al.* [11] as extension of the area under the receiver operating characteristic curve (AUC) [12] to the case of right censored survival data. The AUC is an established performance index for diagnostic tests, which is also a natural indicator of discrimination for binary models [13]. It considers the 'concordance' of the ranking between the predicted probabilities of being 'diseased', for pairs of 'diseased'–'non-diseased' subjects. Similarly, the C index considers the 'concordance' between the ranking of the predicted failure times and that of the observed times, for pairs of subjects [14]. Although the original definition of C would require the prediction of individual survival times, it can be equivalently defined without this need, depending on the model under analysis. Namely, for a proportional hazards model, and for the more general class of transformation models [15], the ranking between the predicted times is equivalent to the opposite ranking of the linear predictor values [11]. More generally, if there is a 'one-to-one' correspondence between predicted times and predicted survival functions, the ranking between predicted times can be obtained by the opposite ranking of the predicted survival probabilities at any fixed time point [10]. Since an increasing number of available covariates (e.g. genetic and molecular characteristics) and long follow-up periods are available in several diseases, models may become rather complex, including covariates having time-dependent effects. In the latter case, as well as in the presence of time-dependent covariates, the 'one-to-one' correspondence may be no longer present. In this case, the use of the original definition of C would require the prediction of individual survival times, e.g. the median or other quantiles, which however may be unavailable due to high censoring. Moreover, this would lead to a measure that may depend on the choice about how to predict the survival times [16].

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Another key aspect concerns the use of C in practice. Applications show the sampling equivalent of C reported in a descriptive way, rather than considering it as an estimate of a population index. In such a case, the sampling variability should be taken into account by providing a confidence interval estimation of C, as proposed by Pencina and D'Agostino [17].

The goal of this work is the evaluation of the discrimination ability of a survival model in a general context, where the 'one-to-one' correspondence may be not present, accounting for the sampling variability. The C^{td} time-dependent discrimination index is proposed as an extension of the Harrell's C which uses the whole predicted survival function as outcome prediction. The index derivation is consistent with the interpretation of the Harrell's C as a weighted average over time of time-dependent accuracy measures [18]. The ability of the predicted survival probability in discriminating patients who develop the event from patients who are still free from the event is considered at any time point, resorting to the concept of time-dependent AUC [18]. The time-dependent AUCs are then summarized dynamically over time, leading to an overall index of discrimination.

The present work was motivated from the practical problem of evaluating the discrimination of a non-proportional hazards model for breast cancer patients. For women with estrogen receptor positive (ER+), tamoxifen is the standard treatment. However, patients may have heterogeneous outcome including a high risk of relapse. Thus, to plan the clinical strategy, models to predict the risk of disease recurrence after adjuvant tamoxifen therapy are needed to support the medical decision. In breast cancer pathology, due to the presence of time-dependent effects of the T pathological stage and hormone receptor contents [19, 20], non-proportional hazards models are needed to account for the prognostic role of covariates. Thus, when assessing the model discrimination ability, the possible absence of the 'one-to-one' correspondence has to be accounted.

In Section 2 (Methods) the ROC methodology and the Harrell's C index for proportional hazards models are briefly described. Then, the meaning of the 'one-to-one' assumption is examined. Finally, the C^{td} index is presented and a non-parametric estimator of C^{td} with an asymptotic confidence interval estimator are provided. In Section 3, the motivating example is presented. Using the C^{td} index, the model discrimination on the training data set is assessed. Model transportability is investigated resorting to an independent testing data set. The paper ends by a discussion in Section 4.

2. METHODS

2.1. Basics on the ROC methodology

Let us consider the binary response setting. D is the disease status indicator, where $D=1$ or 0 depending on whether the subject developed or not an event of interest within a given period of time. Let the dependence between a vector covariates $\mathbf{X}$ and the condition of being 'diseased' be specified through a given binary model (e.g. logistic regression). $z(\mathbf{X})$ is the linear predictor, where greater values of $z(\mathbf{X})$ indicate greater predicted probability of being 'diseased'. The model predictor can be thought as a diagnostic test [13], any value v of $z(\mathbf{X})$ defines a prediction rule to classify a subject as 'diseased' (if $z(\mathbf{X})>v$), or as 'non-diseased' (if $z(\mathbf{X}) \leqslant v$). The probabilities of correct classification conditional on the disease status, i.e. sensitivity $\Pr\{z(\mathbf{X})>v \mid D=1\}$ and specificity $\Pr\{z(\mathbf{X}) \leqslant v \mid D=0\}$ at v, represent

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the predictive accuracy of the rule. The ROC curve, i.e. the plot of sensitivity against the complement to one of the specificity (varying v), represents the accuracy of the whole set of rules. The higher the ROC curve is in the unit square quadrant, the more accurate the prediction rules are. The area under the curve (AUC) constitutes a summary accuracy index, which has been proved to be equal to

$$\mathrm{AUC} = \Pr\{z(\mathbf{X}_i) > z(\mathbf{X}_j) \mid D_i = 1 \ \& \ D_j = 0\}$$

where $\{i, j\}$ denote any pair of randomly chosen subjects [21], and & denotes the intersection operator. Thus, the AUC is the probability that given any pair of 'comparable' subjects, such as their status can be ranked (i.e. one is a 'diseased' and the other one is 'non-diseased'), the ranking between the linear predictor values is 'concordant' with the ranking between the subjects' disease status. $\mathrm{AUC} = 1$ indicates maximum discrimination and $\mathrm{AUC} = 0.5$ indicates absence of discrimination.

Assume a random sample of $n+m$ subjects, where n are 'diseased' (indexed by $i = 1, \ldots, n$) and m are 'non-diseased' (indexed by $j = n+1, \ldots, n+m$), and a binary model, are given. Let $z(\mathbf{x}_1), \ldots, z(\mathbf{x}_{n+m})$ be the observed linear predictor values. For $i = 1, \ldots, n$ $j = n+1, \ldots, m$, let us define the binary indicator $\mathrm{conc}_{ij} = I\{z(\mathbf{x}_i) > z(\mathbf{x}_j)\}$, that is equal to 1 if $\{i, j\}$ are 'concordant', and 0 otherwise. The natural estimator of AUC is the fraction of 'concordant' pairs over the total number of 'comparable' pairs in the sample [22]

$$\widehat{\mathrm{AUC}} = \frac{\sum_{i=1}^{n} \sum_{j=n+1}^{n+m} \mathrm{conc}_{ij}}{n \cdot m}$$

which is equivalent to the Mann–Whitney U-statistic [23].

2.2. The Harrell's C discrimination index

Let us consider the survival time setting, where (T, D) denote the random variables of concern. $T = \min(T_\mathrm{s}, T_\mathrm{c})$ is the observed time, T_s is the failure time, i.e. the time elapsed from $t_{(0)}$ (e.g. time of diagnosis, start of a treatment) to the event of interest; and T_c is the right censoring time, i.e. the time elapsed from $t_{(0)}$ to the end of the study period or the loss to follow-up. It is a matter of fact that in practical applications a discrete time scale is considered (days, weeks, months, etc.). T can be considered as a discrete variable, that assumes values $t_{(1)} < \cdots < t_{(K)}$ (enumerated by $k = 1, \ldots, K$), where $t_{(K)}$ is time elapsed from $t_{(0)}$ to the end of the study period. D is the disease status indicator, $D = 1$ if $T = T_\mathrm{s}$ and $D = 0$ if $T = T_\mathrm{c}$.

Let the dependence between a vector of fixed covariates $\mathbf{X}$ and T_s be specified through a given survival model. In the absence of censoring, for any pair of subjects $\{i, j\}$, labelled such as $T_{\mathrm{s}i} < T_{\mathrm{s}j}$, a desirable property of the model is to have the same ranking between the corresponding predicted times. In such a case, the pair is said to be 'concordant', and the probability of 'concordance' is a 'natural' indicator of model discrimination. In the light of this idea, the C discrimination index considers the probability of 'concordance' addressing for the presence of right censored times. It originated from a modification of the Kendall–Goodman–Kruskal–Somers type rank correlation index [24] and was motivated by a modification of Kendall's τ by Brown [14] and Schemper [25]. The starting point to define whether $\{i, j\}$ are 'concordant' is that they are 'comparable', meaning their survival times $T_{\mathrm{s}i}$ and $T_{\mathrm{s}j}$ can be ranked. The ranking between $T_{\mathrm{s}i}$ and $T_{\mathrm{s}j}$ can be determined from the observed data $(T_i; D_i)$ and $(T_j; D_j)$ if and only if the minimum between T_i and T_j is an event time, i.e. if $T_i < T_j$ and

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$D_i = 1$, or $T_i > T_j$ and $D_j = 1$. Thus, assuming the subjects are labelled such as $T_i < T_j$, they are 'comparable' if and only if $D_i = 1$, arguing that $T_{si} < T_{sj}$. The probability that $\{i, j\}$ are 'comparable' is

$$\pi_{\text{comp}} = \Pr\{T_i < T_j \ \& \ D_i = 1\} \tag{1}$$

where the presence of censored times is considered as a population characteristic and is included in the definition of π_{comp}. If $T_i = T_j$, the pair is considered 'comparable' if i developed the event (i.e. $D_i = 1$) and j was censored the same time (i.e. $D_j = 0$), assuming by convention $T_i < T_j$, and arguing that $T_{si} < T_{sj}$.

2.2.1. Harrell's C for proportional hazards models. In the case of a proportional hazards model, i.e. $\log\{h(t_{(k)} \mid \mathbf{X})\} = z(\mathbf{X}) + \log\{h_o(t_{(k)})\}$ (for $k = 1, \ldots, K$), where $h(t_{(k)} \mid \mathbf{X})$ is the hazard of a patient having covariate equal to $\mathbf{X}$, $h_o(t_{(k)})$ is baseline hazard and $z(\mathbf{X})$ the model linear predictor, C can be defined without the need of predicting individual survival times [10]. The ranking between the predicted times is obtained resorting to the linear predictors, since the inequality $z(\mathbf{X}_i) > z(\mathbf{X}_j)$ is equivalent to $S(t_{(k)} \mid \mathbf{X}_i) < S(t_{(k)} \mid \mathbf{X}_j)$ for any $t_{(k)}$, and this necessarily implies a smaller predicted time for the subject i. Thus, in the absence of censoring, any pair $\{i, j\}$, labelled such as $T_{si} < T_{sj}$, is 'concordant' if and only if $z(\mathbf{X}_i) > z(\mathbf{X}_j)$. In the presence censoring, the C index is the probability that $\{i, j\}$ are 'concordant' given that they are 'comparable'

$$C = \Pr\{z(\mathbf{X}_i) > z(\mathbf{X}_j) \mid T_i < T_j \ \& \ D_i = 1\} = \frac{\pi_{\text{conc}}}{\pi_{\text{comp}}} \tag{2}$$

where the numerator is the probability 'concordance'

$$\pi_{\text{conc}} = \Pr\{z(\mathbf{X}_i) > z(\mathbf{X}_j) \ \& \ T_i < T_j \ \& \ D_i = 1\}$$

and the denominator π_{comp} was defined in (1). In analogy with AUC, C ranges from 0.5 to 1 and has the same interpretation.

Assume a random sample of n subjects and a proportional hazards model are given. $(t_1, d_1, z(\mathbf{x}_1)), \ldots, (t_n, d_n, z(\mathbf{x}_n))$ are the observed times, status indicators and linear predictor values. For $i; j = 1, \ldots, n \ i \neq j$, let us define the binary indicators:

$$\text{comp}_{ij} = I\{t_i < t_j \ \& \ d_i = 1\} + I\{t_i = t_j \ \& \ d_i = 1 \ \& \ d_j = 0\}$$

and

$$\text{conc}_{ij} = I\{z(\mathbf{x}_i) > z(\mathbf{x}_j)\} \cdot \text{comp}_{ij} \tag{3}$$

where $\text{comp}_{ij} + \text{comp}_{ji}$ is equal to 1 if $\{i, j\}$ are 'comparable' and equal to 0 otherwise; and similarly $\text{conc}_{ij} + \text{conc}_{ji}$ is equal to 1 if $\{i, j\}$ are 'concordant' and equal to 0 otherwise. The 'natural' estimator of C, known as the c statistic [11], is the fraction of 'concordant' pairs over the total number of 'comparable' pairs in the sample

$$c = \frac{\sum_{i=1}^{n-1} \sum_{j=i+1}^{n} (\text{conc}_{ij} + \text{conc}_{ji})}{\sum_{i=1}^{n-1} \sum_{j=i+1}^{n} (\text{comp}_{ij} + \text{comp}_{ji})} = \frac{\sum_{i=1}^{n} \sum_{j=1; j \neq i}^{n} \text{conc}_{ij}}{\sum_{i=1}^{n} \sum_{j=1; j \neq i}^{n} \text{comp}_{ij}} \tag{4}$$

A confidence interval estimator for C has been proposed resorting to the theory of U-statistics [26] and to asymptotic results of the modified Kendall's tau index [17]. To correct

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for the optimism arising when C is estimated on the training data set, a bootstrap resampling strategy have been proposed [10].

2.2.2. On the meaning of the 'one-to-one' correspondence. A further definition of C was given for a survival model which specifies the relation between a vector of possibly time-dependent covariates $\mathbf{X}(\mathbf{t})$ (where $\mathbf{t}$ denotes the time instants where there are covariate variations), and T_s, originating a 'one-to-one' correspondence between predicted times and predicted survival probabilities at any time point t [10]. This is guaranteed for example for proportional hazards models (over the entire reference follow-up time), and for the more general class of transformation models [15]. It has to be pointed out, as in the absence of a specification about how to predict individual survival times, the ranking between the predicted times of any $\{i,j\}$, can be obtained from the ranking between $S(t\,|\,\mathbf{X}_i(\mathbf{t}))$ and $S(t\,|\,\mathbf{X}_j(\mathbf{t}))$, if and only if the survival functions are separated over the whole follow-up. Thus, i has lower predicted time than j if and only if $S(t\,|\,\mathbf{X}_i(\mathbf{t}))<S(t\,|\,\mathbf{X}_j(\mathbf{t}))$ for any t. Therefore, the probability of 'concordance' becomes

$$\pi_{\text{conc}} = \Pr\{S(t\,|\,\mathbf{X}_i(\mathbf{t}))<S(t\,|\,\mathbf{X}_j(\mathbf{t}))\ \&\ T_i<T_j;\ D_i=1\}$$

and the C index becomes

$$C = \frac{\pi_{\text{conc}}}{\pi_{\text{comp}}} = \Pr\{S(t\,|\,\mathbf{X}_i(\mathbf{t}))<S(t\,|\,\mathbf{X}_j(\mathbf{t}))\,|\,T_i<T_j\ \&\ D_i=1\} \tag{5}$$

where π_{comp} is defined in (1). The 'natural' estimator of C is obtained substituting $S(t\,|\,\mathbf{X}(\mathbf{t}))$ in place of the linear predictor in (3) and inverting the inequality between the linear predictors.

2.3. The time-dependent discrimination index

We assume of dealing with a survival model in a general framework where the 'one-to-one' correspondence does not necessarily hold. A basic notation is introduced referring to discrete times. For any $t_{(k)}$, $D(t_{(k)})$ denote the 'status at $t_{(k)}$', where $D(t_{(k)})=1$ if the subject is 'diseased at $t_{(k)}$', $D(t_{(k)})=0$ if the subject is 'non-diseased till $t_{(k)}$', and $D(t_{(k)})$ is not defined if the subject is not 'at risk' at $t_{(k)}$. To evaluate the model ability to discriminate among the subjects 'at risk' at $t_{(k)}$, between 'diseased at $t_{(k)}$' and 'non-diseased till $t_{(k)}$', the predicted survival probability at $t_{(k)}$ ($S(t_{(k)}\,|\,\mathbf{X}(\mathbf{t}))$) is the 'natural' quantity to consider. Any value v in $[0,1]$ defines a prediction rule to classify subjects at risk at $t_{(k)}$, as 'diseased at $t_{(k)}$' (if $S(t_{(k)}\,|\,\mathbf{X}(\mathbf{t}))\leqslant v$), or 'not diseased till $t_{(k)}$' (if $S(t_{(k)}\,|\,\mathbf{X}(\mathbf{t}))>v$). The probabilities of correct classification conditional on the 'status at $t_{(k)}$', i.e. incident sensitivity $\Pr\{S(t_{(k)}\,|\,\mathbf{X}(\mathbf{t}))\leqslant v\,|\,D(t_{(k)})=1\}$ and dynamic specificity $\Pr\{S(t_{(k)}\,|\,\mathbf{X}(\mathbf{t}))>v\,|\,D(t_{(k)})=0\}$ [18, 27], represent the accuracy of the rule, and the area $\text{AUC}(t_{(k)})$ under the corresponding ROC curve

$$\text{AUC}(t_{(k)}) = \Pr\{S(t_{(k)}\,|\,\mathbf{X}_i(\mathbf{t}))<S(t_{(k)}\,|\,\mathbf{X}_j(\mathbf{t}))\,|\,D_i(t_{(k)})=1\ \&\ D_j(t_{(k)})=0\} \tag{6}$$

represents the discrimination ability of $S(t_{(k)},\mathbf{X}(\mathbf{t}))$. To summarize the ability of $S(t_{(k)},\mathbf{X}(\mathbf{t}))$ to discriminate between $D(t_{(k)})=1$ and $D(t_{(k)})=0$ over the whole follow-up, we propose the C^{td} index as a weighted average of $\text{AUC}(t_{(k)})$ over time

$$C^{\text{td}} = \frac{\sum_{k=0}^{K}\text{AUC}(t_{(k)})\cdot w(t_{(k)})}{\sum_{k=0}^{K} w(t_{(k)})} \tag{7}$$

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where

$$w(t_{(k)}) = \Pr\{D_i(t_{(k)}) = 1 \ \& \ D_j(t_{(k)}) = 0\} \tag{8}$$

is the probability that $\{i, j\}$ are 'comparable at $t_{(k)}$', i.e. one subject is 'diseased at $t_{(k)}$' and the other one is 'non-diseased till $t_{(k)}$'. It is worth of note as $w(t_{(k)})$ is the same weighting system adopted by Heagerthy and Zheng [18] in their derivation of the Harrell's C as weighted average over time of time-dependent accuracy. According to the definition of the Harrell's C censoring is included as a population characteristic.

The denominator of (7) becomes

$$\sum_{k=0}^{K} w(t_{(k)}) = \pi_{\text{comp}} \tag{9}$$

where π_{comp} was defined in (1), and the numerator (7) becomes

$$\pi_{\text{conc}}^{\text{td}} = \sum_{k=0}^{K} \text{AUC}(t_{(k)}) \cdot w(t_{(k)}) = \Pr\{S(T_i \mid \mathbf{X}_i(\mathbf{t})) < S(T_i \mid \mathbf{X}_j(\mathbf{t})) \ \& \ T_i < T_j \ \& \ D_i = 1\} \tag{10}$$

The proofs of (9) and (10) are provided in Appendix A (1 and 2). Substituting (9) and (10) in (7), it follows

$$C^{\text{td}} = \frac{\pi_{\text{conc}}^{\text{td}}}{\pi_{\text{comp}}} = \Pr\{S(T_i \mid \mathbf{X}_i(\mathbf{t})) < S(T_i \mid \mathbf{X}_j(\mathbf{t})) \mid T_i < T_j \ \& \ D_i = 1\} \tag{11}$$

Let us observe as the difference between C (5) and C^{td} is that in the latter the predicted survival is evaluated in T_i, which indeed depends on $\{i, j\}$, instead on a fixed t. C^{td} is the probability that $\{i, j\}$ are 'concordant' given they are 'comparable', where the 'concordance' is defined on the ground of the following consideration. Given a 'comparable' pair $\{i, j\}$, labelled such as $T_i < T_j$ and $D_i = 1$, a desirable property of the model is that the predicted survival probability, at the time where the subject i developed the event, is greater for the subject j who actually is still free from the event. We refer this condition as 'td concordance'.

If the 'one-to-one' correspondence holds, C^{td} reduces to the Harrell's C. In fact, in this case the inequality $S(T_i \mid \mathbf{X}_i(\mathbf{t})) < S(T_i \mid \mathbf{X}_j(\mathbf{t}))$ holds if and only if $S(t_{(k)} \mid \mathbf{X}_i(\mathbf{t})) < S(t_{(k)} \mid \mathbf{X}_j(\mathbf{t}))$ for any $t_{(k)}$, or equivalently if $S(t \mid \mathbf{X}_i(\mathbf{t})) < S(t \mid \mathbf{X}_j(\mathbf{t}))$ for any fixed t. This implies the equality between π_{conc} and $\pi_{\text{conc}}^{\text{td}}$, and being π_{comp} the denominator common to both C^{td} and C, the equality $C^{\text{td}} = C$ follows. In practice, we would typically restrict the attention to a fixed follow-up period of time $[t_{(0)}; \tau]$ (e.g. 5 years), where $\tau < t_{(K)}$, scanning in (7) only the times $t_{(k)} \leqslant \tau$. This leads to an index corresponding to (11) after having introduced administrative censoring at τ.

2.3.1. Point and interval estimation of C^{td}. Assume a random sample of n subjects and a survival model are given. $(t_1, d_1, S(t_{(k)} \mid \mathbf{x}_1(\mathbf{t}));\ k = 1, \ldots, K), \ldots, (t_n, d_n, S(t_{(k)} \mid \mathbf{x}_n(\mathbf{t}));\ k = 1, \ldots, K)$ are the observed times, status indicators and predicted survival function. For $i;\ j = 1, \ldots, n\ i \neq j$, the binary indicator

$$\text{conc}_{ij}^{\text{td}} = I\{S(t_i \mid \mathbf{x}_i(\mathbf{t})) < S(t_j \mid \mathbf{x}_j(\mathbf{t}))\} \cdot \text{comp}_{ij} \tag{12}$$

is defined in analogy with (3). $\text{conc}_{ij}^{\text{td}} + \text{conc}_{ji}^{\text{td}}$ is equal to 1 if the pair is 'td concordant' and equal to 0 otherwise. The 'natural' estimator of C^{td} is the fraction of 'concordant' pairs over

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the total number of 'comparable' (according to the previously given definition) pairs in the sample

$$c^{\mathrm{td}} = \frac{\sum_{i=1}^{n} \sum_{j=1;\, j\neq i}^{n} \mathrm{conc}_{ij}^{\mathrm{td}}}{\sum_{i=1}^{n} \sum_{j=1;\, j\neq i}^{n} \mathrm{comp}_{ij}} \tag{13}$$

Let us observe as c^{td} can be thought as the ratio of the estimates of $\pi_{\mathrm{comp}}^{\mathrm{td}}$ and π_{conc} obtained from the two correlated one-sample U-statistics

$$u_{\mathrm{conc}}^{\mathrm{td}} = \frac{2}{n(n-1)} \sum_{i=1}^{n-1} \sum_{j=i+i}^{n} \frac{1}{2}(\mathrm{conc}_{ij}^{\mathrm{td}} + \mathrm{conc}_{ji}^{\mathrm{td}}) \tag{14}$$

$$u_{\mathrm{comp}} = \frac{2}{n(n-1)} \sum_{i=1}^{n-1} \sum_{j=i+i}^{n} \frac{1}{2}(\mathrm{comp}_{ij} + \mathrm{comp}_{ji}) \tag{15}$$

Owing to this interpretation, the jackknife version $c_{\mathrm{jack}}^{\mathrm{td}}$ of c^{td} can be considered to derive a confidence interval for C^{td}. To introduce $c_{\mathrm{jack}}^{\mathrm{td}}$ we need to define the following notation. For $i = 1, \ldots, n$, let $c^{\mathrm{td},-i} = u_{\mathrm{conc}}^{\mathrm{td},-i} / u_{\mathrm{comp}}^{-i}$, where $u_{\mathrm{conc}}^{\mathrm{td},-i}$ and u_{comp}^{-i} are equivalents (14) and (15) evaluated on $n-1$ sampling observations after excluding the subject i, and let $c^{\mathrm{td},i} = n \cdot c^{\mathrm{td}} - (n-1) \cdot c^{\mathrm{td},-i}$ be the Tuckey's pseudo-values of c^{td}. The jackknife version of c^{td} is

$$c_{\mathrm{jack}}^{\mathrm{td}} = \frac{1}{n} \sum_{i=1}^{n} c^{\mathrm{td},i} = c^{\mathrm{td}} - \frac{n-1}{n} \sum_{i=1}^{n} c^{\mathrm{td},-i}$$

and the jackknife sampling variance is

$$s_{\mathrm{jack}}^{2\,\mathrm{td}} = \frac{1}{n-1} \sum_{i=1}^{n} (c^{\mathrm{td},i} - c_{\mathrm{jack}}^{\mathrm{td}})^2 \tag{16}$$

Let us observe as in the calculation of the pseudo-values the model was not refitted for each 'leave one out' sample, thus the variance of the estimated regression coefficients is not taken into account in (16). However, by refitting the model for each 'leave one out' sample, overfitting can be taken into account. By using Arversen's [28] contributions on the asymptotic behaviour of the jackknife version of statistics obtained as functions of correlated U-statistics satisfying some regularities, it follows that $c_{\mathrm{jack}}^{\mathrm{td}}$ is asymptotically unbiased for C_{td} and normally distributed. Moreover (16) is a consistent estimator of the asymptotic variance of $c_{\mathrm{jack}}^{\mathrm{td}}$. Thus, as an asymptotic $(1-\alpha)$ per cent confidence interval for C_{td} is

$$[c_{\mathrm{jack}}^{\mathrm{td}} \pm z_{1-\alpha/2} \cdot s_{\mathrm{jack}}^{\mathrm{td}} / \sqrt{n}]$$

where $z_{1-\alpha/2}$ is the percentile of the standard normal distribution. The methodology employed has been utilized in previous work on non-parametric inference on difference of correlated Harrell's C [26] and can be generalized to the case of the comparison of correlated C^{td} indexes. A confidence interval for a single C^{td} could also be obtained from the unjackknifed estimator (13). The asymptotic variance can be estimated by its sampling equivalent, in the same fashion of the confidence interval for the Harrell's C provided in by the rcorr.cens function in the R/S-Plus Hmisc library.

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3. MOTIVATING EXAMPLE: A NON PROPORTIONAL HAZARDS MODEL FOR BREAST CANCER PATIENTS

In the planning of adjuvant treatment in breast cancer, steroid receptors provide the main criterion for identifying patients which would benefit from hormonal therapy. For postmenopausal patients with axillary lymph-node involvement (N+), although tamoxifen is the standard treatment for estrogen receptor positive (ER+) breast cancer, patients may have not complete response to treatment showing disease recurrence even within the treatment period. As an aid to plan the addition of a more aggressive therapy to tamoxifen, a model to predict the risk of disease recurrence after adjuvant tamoxifen was developed from a case series provided by the Istituto Nazionale per lo Studio e la Cura dei Tumori di Milano (INT). Considering that a basic requirement a model has to satisfy to be proposed for clinical use is transportability, the predictive accuracy of the model was then assessed on independent testing data set, provided by Centro Regionale Indicatori Biochimici di Tumore (CRIBT) in Venice.

3.1. Training case series (*INT*)

A cohort of 596 consecutive breast cancer patients, who entered prospectively at Istituto Nazionale per lo Studio e la Cura dei Tumori di Milano (INT), from 1984 to 1996, was considered. Patients underwent radical or conservative surgery followed by radiotherapy and were submitted to adjuvant tamoxifen (planned for at least 36 months, started within 4 weeks from the surgery). The selection criteria were: absence of previous primary chemotherapy or adjuvant chemotherapy, absence of synchronous bilateral tumours, absence of previous or concomitant tumours different from breast cancer, absence of relapse within 6 months from the surgery, potential follow-up of at least 60 months, N+, pT_1, pT_2, pT_3, M_0 pathological stage, at least 10 axillary lymph node analysed, where at most 9 are metastatic, lobular, ductal or mixed histology, non-plurifocal tumour, estrogen receptors content of at least 10 fmoli/protein mg (ER+), post-menopause (i.e. amenorrhea since at least 1 year before the surgery). The probability of being under follow-up [29], was equal to 0.75, 0.50, 0.25 at 59, 78 and 95 months, respectively. The endpoint of interest was the time to relapse defined as the first event among: locoregional relapse, distant metastasis and contralateral tumours. The observed events were: 27 locoregional relapses, 109 distant metastasis, 18 contralateral tumours and 13 new primary tumours different from breast cancer. The statistical analysis was based on a reference time period of 60 months with 111 events.

3.2. Testing case series (*CRIBT*)

A cohort of 175 patients, who underwent radical or conservative surgery followed by radiotherapy and were submitted to adjuvant tamoxifen therapy (planned for at least 36 months), was considered. Tumour tissues were consecutively sent to the Centro Regionale Indicatori Biochimici di Tumore (CRIBT) in Venice, from 1982 to 1992. The selection criteria adopted are the same used for the INT case series. The probability of being under follow-up was equal to 0.75, 0.50, 0.25 at 51, 64 and 80 months, respectively. The observed events were: 9 loco regional relapses, 48 distant metastasis, 1 contralateral tumours. Two patients with neoplastic event are unknown for the site of relapse. The endpoint considered was the same of the INT cohort.

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3.3. The model

The prognostic impact of estrogens (ER) and progesterone (PgR) receptor contents, number of metastatic axillary lymph nodes (N), T stage (T) and histology (Hist) on the development of relapse was considered. A piecewise exponential regression model, involving the partition of the time axis into intervals, was used. The prognostic effect of the covariates on time to relapse were included accounting for previous knowledge [20]. A time-dependent effect for the pathological T stage was included by interaction terms with time leading to a right continuous step function constant within each interval [30]. A time-dependent effect for the estrogens receptors contents was not included, as in previous work evidences of time-effects were present only after 5 years of follow-up. The choice of a piecewise exponential model, motivated from intended use for predictive purpose, enabled to obtain a fully parametric estimate of the shape of the hazard as a function of time and covariates, avoiding restrictive assumptions on the distribution of time to relapse. Such an approach was proposed in the more general framework of cause-specific hazards for competing risks [19]. It is worth noticing as in the presence of proportional hazards, the estimates of the covariate effects obtained from a piecewise exponential model are equivalent to those obtained from a Cox proportional hazards model [31].

According to the monthly time scale of the observed data, the follow-up time was partitioned into $L=60$ disjoint monthly intervals $A_l=(l-1;l]$ $(l=1,\ldots,L)$. The prognostic effect of the covariates $(\mathbf{X}(\mathbf{t}))$ on time to relapse, for $t\in A_l$, is

$$\begin{aligned}\log\{h(t,\mathbf{X}(\mathbf{t}))\} = {} & \beta_0+\beta_1\cdot l_1+\beta_2\cdot l_2+\beta_3\cdot l_3\}z(l)\\ & \left.\begin{aligned}&+\beta_4\cdot \mathrm{ER}_1+\beta_5\cdot \mathrm{ER}_2+\beta_6\cdot \mathrm{PgR}+\beta_7\cdot N+\\ &+\beta_8\cdot \mathrm{Hist}_1+\beta_9\cdot \mathrm{Hist}_2\end{aligned}\right\}u(\mathbf{X})\\ & +\beta_7\cdot T+\beta_8(T\cdot l_1)+\beta_9(T\cdot l_2)+\beta_{10}(T\cdot l_3)\}v(T;l)\end{aligned}$$

where $h(t,\mathbf{X}(\mathbf{t}))$ is the hazard of relapse (constant for $t\in A_l$), l_1 (linear effect of time) l_2, l_3 (non-linear effects of time) are the basis of the four-knots $(12,24,37,56)$ restricted cubic spline on the midpoints l of the time intervals A_l. ER_1 and ER_2 (linear and non-linear effects of ER) are the basis of the three-knots $(2.9,4.8,6.3)$ restricted cubic spline on log(ER). PgR is the linear effect of log(PgR). Hist_1 and Hist_2 are two dummy variables to code Hist ($\mathrm{Hist}_1=1$ if lobular and $\mathrm{Hist}_1=0$ otherwise; $\mathrm{Hist}_2=1$ if mixed and $\mathrm{Hist}_2=0$ otherwise). T is a dummy variable to code T ($T=0$ if pT_1, $T=1$ if pT_2 or pT_3). $T\cdot l_1$, $T\cdot l_2$, $T\cdot l_3$ are the interactions between T stage and time. Restricted cubic splines were obtained using the rcs function in S-PLUS 2000 [32] Design library.

In Table I the results of the model fitting are showed. In addition to the coefficient's estimates, Wald statistic is also reported for the contribution of each variable.

To investigate whether the 'one-to-one' correspondence holds, the hazard functions pertaining to minimum, maximum and the deciles values of the empirical distribution of $u(\mathbf{X})$ in pT_1 and $\mathrm{pT}_2/\mathrm{pT}_3$ patients ($u^1_1,\ldots,u^1_{11}$, and $u^{23}_1,\ldots,u^{23}_{11}$) were calculated. The corresponding survival curves were plotted (see Figure 1). The survival function for in pT_1 patients was obtained, for $t\in A_l$ and $q=1,\ldots,11$, by

$$S(t\mid u^1_q)=\exp\{-H(t\mid u^1_q)\} \tag{17}$$

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Table I. Model results.

Variable	b	es(b)	χ^2	d.f.	p-value
Intercept	−4.867	1.047	21.589	1	<0.001
t_1	0.072	0.040	3.211	1	0.073
t_2	−0.040	0.191	0.044	1	0.834
t_3	−0.172	0.511	0.114	1	0.736
t	—	—	12.752	3	0.005
ER_1	−0.680	0.212	10.243	1	0.001
ER_2	0.684	0.221	9.554	1	0.002
ER	—	—	10.373	2	0.006
PgR	−0.141	0.039	12.851	1	<0.001
N	0.803	0.131	37.524	1	<0.001
$Hist_1$	0.287	0.260	1.217	1	0.270
$Hist_2$	−0.087	0.354	0.060	1	0.806
Hist	—	—	1.226	2	0.542
T	−0.336	0.927	0.131	1	0.717
$t_1 \cdot T$	0.070	0.0537	1.698	1	0.192
$t_2 \cdot T$	−0.636	0.251	6.398	1	0.011
$t_3 \cdot T$	1.930	0.664	8.438	1	0.004
td(T)	—	—	12.874	3	0.005
T	—	—	15.237	4	0.004

Estimated regression coefficients (b), standard error (es), Wald statistic (χ^2) obtained by the model on the INT case series. t is the overall effect of time, ER is the overall effect of estrogen receptor status, Hist is the effect of tumour histology, td(T) is the time-dependent effect of T stage, and T is the overall effect of pathological T stage.

where

$$H(t \mid u_q^1) = \sum_{m=1}^{l-1} h(m, u_q^1) \cdot (a_m - a_{m-1}) + h(l, u_q^1) \cdot (t - (l-1)) \tag{18}$$

and similarly for pT_2/pT_3 patients. Several crossings are present among the predicted survival curves, thus, the 'one-to-one' correspondence is not verified.

3.4. Model discrimination—training data set

C^{td} was applied to assess the model discrimination on the training data set. For each subject, $S(t, \mathbf{X}(\mathbf{t}))$, at any required t, was calculated according to (17) starting from

$$h(l \mid \mathbf{X}(\mathbf{t})) = \exp\{z(l) + u(\mathbf{X}) + v(T, l)\}$$

S-plus code for calculation of $S(t, \mathbf{X}(\mathbf{t}))$ is provided in Appendix B.

The estimator c^{td} was considered for a reference time τ ranging between the minimum and the maximum observed event times up to 60 months. The behaviour of c^{td} according to τ is shown in Figure 2(a). c^{td} is oscillating till about 17 months ($\min = 0.653, \max = 0.823$), and afterwards becomes stable. At 36, 48, 60 months, c^{td} is equal to 0.724, 0.719, and 0.729, respectively. The oscillation of c^{td} till 17 months was expected, given the percentage of 'comparable' pairs is less or equal than 22 per cent of the total number of 'comparable' pairs. The bootstrap estimated optimism [10] (200 bootstrap samples) is stable around about 0.020

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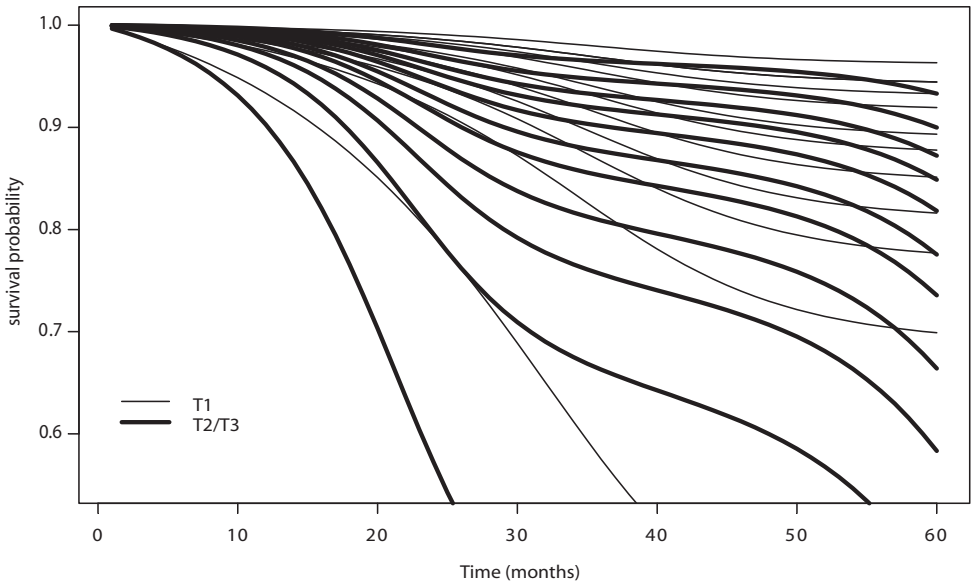


Figure 1. Check of the 'one-to-one' assumption. Plot of $S(t, v_{1,l})$ and $S(t, v_{2,l})$ $(l = 1, \ldots, 11)$ against time. $v_{1,l}$ and $v_{2,l}$ are the minimum, maximum and the deciles of the constant part of the linear predictor, in pT_1 (T1) and pT_2/pT_3 (T2/T3) patients, respectively.

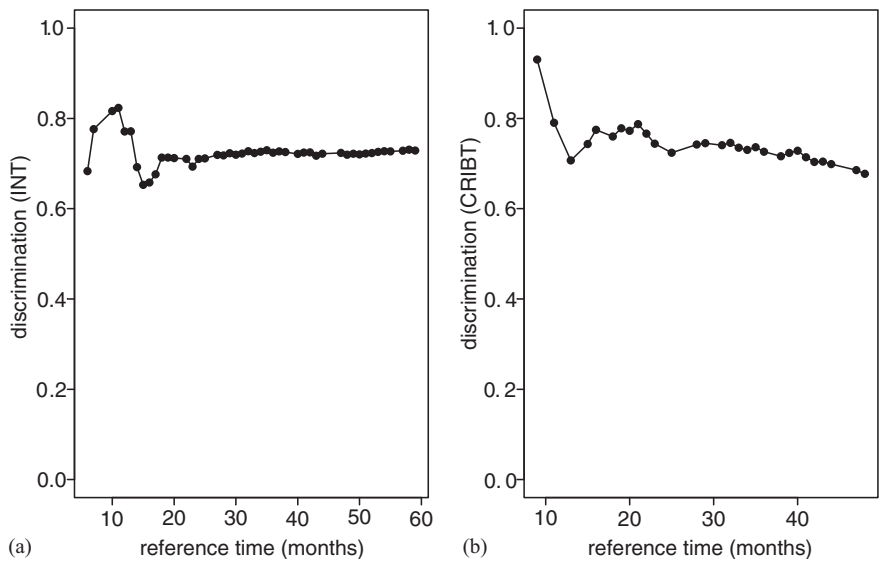


Figure 2. Discrimination performance on the training and testing data sets. Plot of c^{td} against τ: (a) INT data set; and (b) CRIBT data set.

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across the considered reference times. Thus, the bias-corrected assessments of discrimination are 0.704, 0.699 and 0.709, showing a modest shrinkage effect due to overoptimism. The jackknife 95 per cent confidence intervals are [0.671; 0.782], [0.672; 0.772], [0.685; 0.775], respectively.

3.5. *Validation of model discrimination*

At 36, 48 and 60 months the probability of being under follow-up in the testing data set are about 0.93, 0.84 and 0.54, compared to 0.90, 0.83 and 0.74 of the training data set. Thus, the validation of the discrimination ability was based on a maximum reference time $\tau = 48$ months. The behaviour of c^{td} according to τ is shown in Figure 2(b). c^{td} is oscillating till about 20 months ($\min = 0.704, \max = 0.924$), where the percentage 'comparable' pairs is less or equal than 30 per cent of the total. Afterwards c^{td} stabilizes, showing however a slight tendency to decrease over time. At 36 and 48 months, c^{td} is equal to 0.735 (78 per cent of 'comparable' pairs) and 0.687. The jackknife 95 per cent confidence intervals at 36 and 48 months are [0.642; 0.819], [0.601; 0.763], respectively.

Considering the values of c^{td} on the testing data sets obtained at 36 and 48 months, and the estimated optimism, it appears as the model maintains its discriminating performance on the independent testing data set, thus showing a condition for its transportability. The confidence intervals obtained from the training data set are narrower than those obtained from the testing data set. This was expected given the lower number of events in the testing data set. However, it can be noticed as the confidence intervals obtained from the training data set are included in the corresponding ones obtained from the testing data set.

4. DISCUSSION

When model development is intended to derive tools suitable for outcome prediction, a key aspect is the availability of appropriate measures of model predictive accuracy, to evaluate the generalizability to new patients of different prognostic models or to judge the relevance of additional contributions of new covariates. These aspects require that measures have to be evaluable on independent testing data sets, and suitable to a general class of model formulations, given that models may become rather complex (e.g. including non-linear and time-dependent covariate effects).

Among the approaches proposed in the biostatistical literature, discrimination measures own a straightforward interpretability, referring to the ability of outcome predictions to separate subjects consistently to their actual outcome. In this context, an established discrimination measure for binary data is the area under the ROC curve. The C index is its natural extension for right censored survival data, which considers the presence of censoring as a population feature rather than a limitation of the sample. A condition of 'concordance' is in fact defined only for subjects that can be compared on the ground of their observed times and status indicators. For the class of proportional hazards models and for the broader class of models satisfying the 'one-to-one' correspondence (i.e. transformation models), C can be defined from the linear predictor or the predicted survival probability at any fixed time point, respectively. In fact, in the presence of a separation among the model predicted survival functions, any reasonable estimator of the survival times would lead to the same index. If the 'one-to-one'

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correspondence does not hold, for example because of the presence of time-dependent effects of fixed covariates or time-dependent covariates, the definition of C would require the prediction of individual times, which however may be unavailable. Moreover, this would lead to a measure that may depend on the choice about how to predict the survival times.

The proposed C^{td} index is a rank based measure that overcomes the issue of predicting survival times, which can be utilized for a general class of models, even in the absence of the 'one-to-one' correspondence as the whole individual predicted survival function is directly utilized. C^{td} is based on a novel definition of 'concordance' on the ground of a desirable property of the model: a subject who developed the event is expected to have a smaller probability of surviving beyond his/her survival time than any subject who survived longer. C^{td} ranges from 0.5, representing absence of discrimination, and 1 maximum discrimination; and reduces to C in the presence of 'one-to-one' correspondence.

The sampling equivalent of C was considered as 'natural' descriptive index summarizing the discrimination of a given model applied on a sample. A confidence interval estimator was also proposed resorting to Arversen's [28] contributions on the jackknife method applied to functions of correlated one-sample U-statistics.

The usefulness of the proposed measure relies on the evaluation of models on data characterized by long a follow-up where the prognostic effect of covariate may vary over time. This was a characteristic of the model for prediction of risk of relapse in breast cancer patients treated with tamoxifen, that motivated the present work. The presence of a time-dependent effect of the pathological T stage resulted in crossing predicted survival functions. The proposed measure enabled to investigate the discrimination performance this model on its training data set and on an independent testing.

As C^{td} is a discrimination measure having a straightforward interpretability and applicability in a wide range of possibly complex survival models, a deeper investigation on the properties of this measure is desirable. Being a simple extension of the Harrell's C, the index treats censoring the same way as in the original C. Concerning the robustness of C^{td} under different censoring schemes, when considering an underlying uncensored population [33], it is worth noticing as censoring acts on the conditional event of the index, which is the same of C, therefore C^{td} shares necessarily to the same characteristics of the Harrell's C. In a recent investigation on robustness under censoring of C [34], the degree of bias increases as the percentage of censored data increases and appears to be largest for moderate levels of 'true' concordance. However, a deeper investigation on robustness under censoring of C^{td} is needed and will be matter of further work. In addition, alternative weighting schemes which could minimize the effect of censoring on the global measure's expectation could be considered, e.g. inverse probability weighting of censoring. Finally, an open issue is the inferential procedure for the comparison between the C^{td} values obtained from different model on the same case series. This would support the choice among several prognostic models.

APPENDIX A: DERIVATION OF C^{td}

From the definition of 'status at $t_{(k)}$', given in Section 2.3, $D(t_{(k)})=1$ is equivalent to $T=t_{(k)}$ and $D=1$; and $D(t_{(k)})=0$ is equivalent to $t_{(k)}<T$. The weight $w(t_{(k)})$, defined in (8), can be written as

$$w(t_{(k)}) = \Pr\{T_i = t_{(k)} \ \&\ D_i = 1 \ \&\ t_{(k)} < T_j\} \tag{A1}$$

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and from the definition of conditional probability

$$w(t_{(k)}) = \Pr\{T_i < T_j \mid T_i = t_{(k)} \ \&\ D_i = 1\} \cdot \Pr\{T_i = t_{(k)} \ \&\ D_i = 1\} \tag{A2}$$

1: To prove equality (9), from (A2) the summation $\sum_{k=0}^{K} w(t_{(k)})$ becomes

$$\sum_{k=0}^{K} \Pr\{T_i < T_j \mid T_i = t_{(k)} \ \&\ D_i = 1\} \cdot \Pr\{T_i = t_{(k)} \ \&\ D_i = 1\}$$

and considering $t_{(1)}, \ldots, t_{(K)}$ is the whole support of the (sub)distribution of T given by $(T; D = 1)$, it follows

$$\sum_{k=0}^{K} w(t_{(k)}) = \Pr\{T_i < T_j \ \&\ D_i = 1\} = \pi_{\text{comp}} \tag{A3}$$

2: To prove equality (10), let us consider $\text{AUC}(t_{(k)})$, defined in (6). Arguing for $\text{AUC}(t_{(k)})$ similarly as done to derive (A1), we can write

$$\text{AUC}(t_{(k)}) = \Pr\{S(T_i \mid \mathbf{X}_i(\mathbf{t})) < S(T_i \mid \mathbf{X}_j(\mathbf{t})) \mid T_i = t_{(k)} \ \&\ D_i = 1 \ \&\ T_i < T_j\} \tag{A4}$$

and from (A2) with (A4) the summation $\sum_{k=0}^{K} w(t_{(k)}) \cdot \text{AUC}(t_{(k)})$ becomes

$$\sum_{k=0}^{K} \Pr\{S(T_i \mid \mathbf{X}_i(\mathbf{t})) < S(T_i \mid \mathbf{X}_j(\mathbf{t})) \ \&\ T_i < T_j \mid T_i = t_{(k)} \ \&\ D_i = 1\} \cdot \Pr\{T_i = t_{(k)} \ \&\ D_i = 1\}$$

$$= \Pr\{S(T_i \mid \mathbf{X}_i(\mathbf{t})) < S(T_i \mid \mathbf{X}_j(\mathbf{t})) \ \&\ T_i < T_j \ \&\ D_i = 1\} = \pi_{\text{conc}}^{\text{td}}$$

APPENDIX B: S-PLUS CODE FOR OBTAINING $\hat{S}(t, \mathbf{Z})$

We consider the case of a piecewise exponential model with: time modelled as a four-knots restricted cubic spline, one fixed covariate with no time-dependent effects (cov1), one fixed covariate with time-dependent effects (cov2).

```
# Arguments

# L = number of time intervals

# time.extremes = L × 1 vector containing the upper extremes of the time intervals

# lengths.intervals = L × 1 vector containing the lengths of the time intervals,
   i.e. ( c( time.extremes[ 1 : pos ] - c( 0, time.extremes[ 1 : (pos-1) ]))/2

# time.mid.points = L × 1 vector containing the midpoints of the time intervals,
   i.e. ( c( 0, time.extremes[ 1 : ( L - 1 ) ] ) + time.extremes )/2
```

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```
# MODEL COEFFICIENTS
# coeff.intercept = intercept coefficient

# coeff.time1 = coefficient of linear effect of time
# coeff.time2 = coefficient of non-linear effect of time
# coeff.time3 = coefficient of non-linear effect of time

# coeff.cov1.fixed = fixed covariate coefficient

# coeff.cov2.fixed = coefficient of fixed effect of cov1
# coeff.cov2.time1 = coefficient of linear time-dependent effect of cov1
# coeff.cov2.time2 = coefficient of non-linear time-dependent effect of cov1
# coeff.cov2.time3 = coefficient of non-linear time-dependent effect of cov1

# SET t and COVARIATES

> t    < -
> cov1 < -
> cov2 < -

# CALCULATION OF THE LENGTHS OF THE INTERVALS COVERED BY t

> pos < -sum( extremes < t )

> lengths.covered  < - c( length.intervals[ 1: (pos)] , ( t - time.extremes[pos] ) )


# CALCULATION OF THE CUMULATIVE HAZARD

> time.spline < -rcs( time.mid.points, c( " list of 4 nodes " ) )
> time1  < -time.spline[ ,1 ]
> time2 < - time.spline[ ,2 ]
> time3 < - time.spline[ ,3 ]

> u.X < - ( coeff.intercept + coeff.cov1.fixed * cov1 ) * rep( 1, ( pos + 1 ) )

> v.X.l < -rep( coeff.cov2.fixed, ( pos + 1 ) ) +
                  coeff.cov2.time1 * time1[ 1 : ( pos + 1 ) ] +
                  coeff.cov2.time2 * time2[ 1 : ( pos + 1 ) ] +
                  coeff.cov2.time3 * time3[ 1 : ( pos + 1 ) ]

> z.l < -coeff.time1 * time1[ 1 : ( pos + 1 ) ] +
              coeff.time2 * time2[ 1 : ( pos + 1 ) ] +
              coeff.time3 * time3[ 1 : ( pos + 1 ) ]

> haz.l < -exp( u.X + z.l + v.X.l )
```

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```
# CALCULATION OF THE SURVIVAL

>   cum.haz.l < −sum( haz.l * lengths.covered )

>   surv.l < −exp( - cum.haz.l )
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